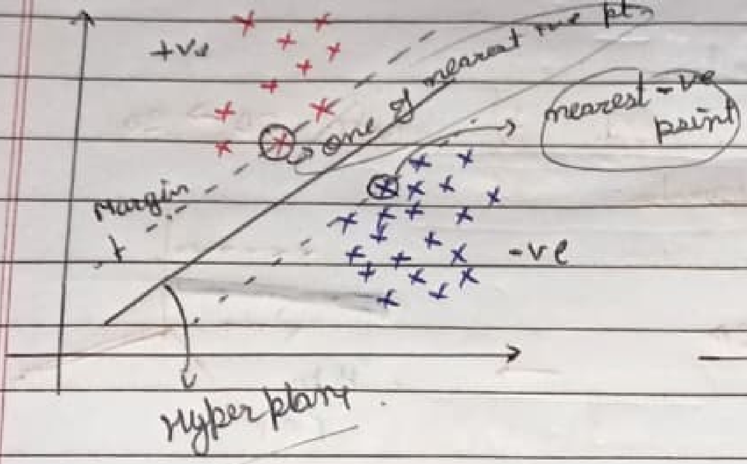
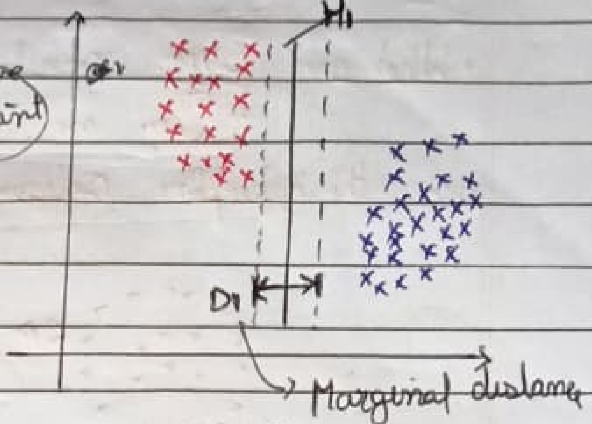
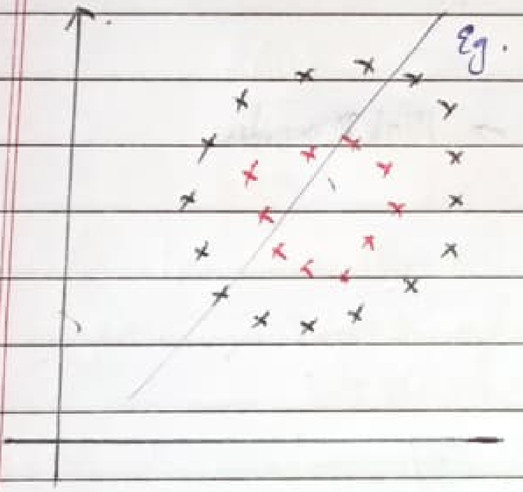
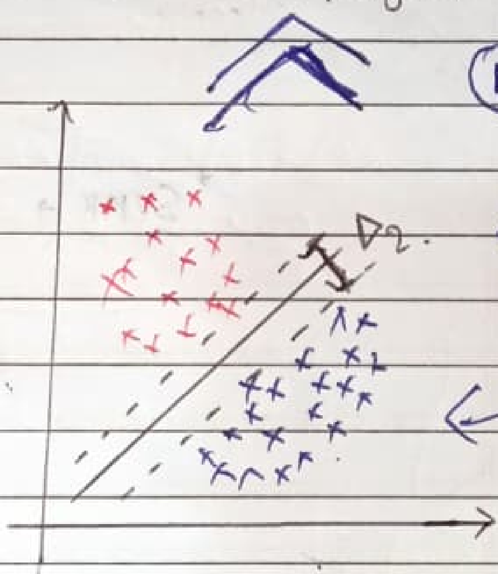
Tutorial-50 { Support Vector Machine }

1. Support Vectors
2. Hyper Planes
3. Marginal distance
4. Linear Separable → On drawing st. line, we can separate → Points
5. Non-linear separables ↳ can't separate points









D2 > D1 ↳ assign

So we consider

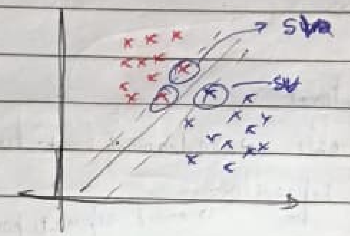
Aim → Marginal distance should be more.
→ make Hyperplane such a way D1 is highest out of all possible D's.

* Marginal HP → +ve → nearest point on +ve side } to the Hyperplane.
  -ve → " " " -ve side }

PTO

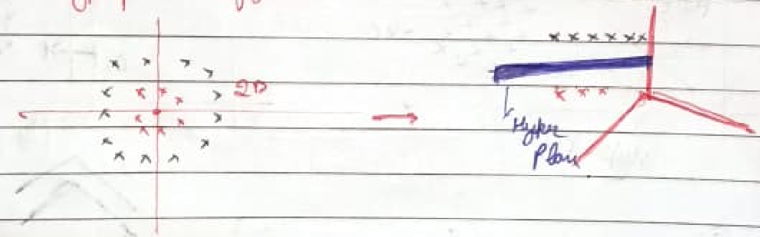
What are Support Vectors

→ The points through which marginal hyperplanes pass.



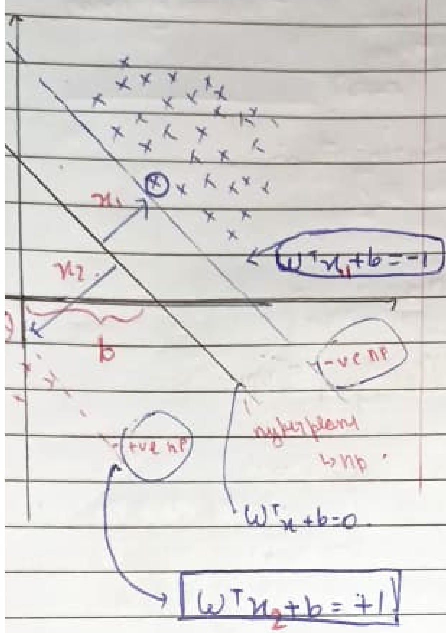
• What are SVM Kernels.

→ They are used to make 2D → 3D a/c the hyperplane configuration.

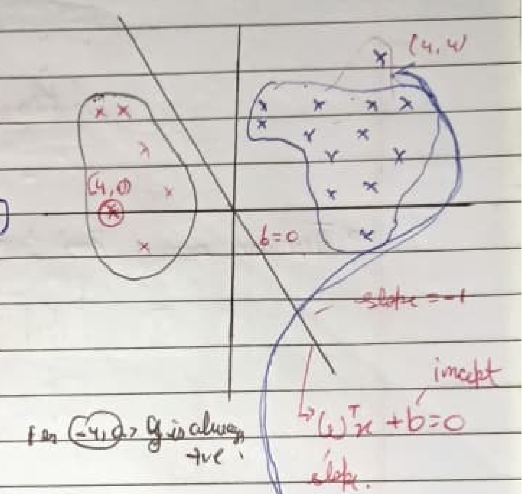


→ SVMK → Low Dim → High Dimensions

SVM math Intution Pt. II.



$$\boxed{w^T x_2 + b = +1}$$



For (-4, 0) y is always +ve

$y = w^T x + b$

$= \begin{bmatrix} -1 \\ 0 \end{bmatrix} \begin{bmatrix} 4 & 4 \end{bmatrix}$

$\Rightarrow -4$ -ve

→ * In SVM, we consider max (marginal Plane)

$(x_2 - x_2)$

$\Rightarrow w_T x_1 + \cancel{b} = -1$

$w_T x_2 + \cancel{b} = +1$

(-) (-) (-)

$w_T (x_1 - x_2) = -2$

$w_T (x_2 - x_1) = 2.$

$$\left\{ \frac{w_T}{\|w\|} (x_2 - x_1) = \frac{2}{\|w\|} \right.$$

Optimization function

$$\boxed{(w^*, b^*) \max \frac{2}{\|w\|} \quad s.t. \quad y_i \begin{cases} +1 & w^T x + b \geq 1 \\ -1 & w^T x + b \leq 1 \end{cases}}$$

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$$(w^*, b^*) = \min \frac{\|w\|}{2} + C_i \sum_{i=1}^{n} \xi_i$$

zeta.

How many errors?

Values of error.

Regularization!!

* These are eq. of Hard SVMs

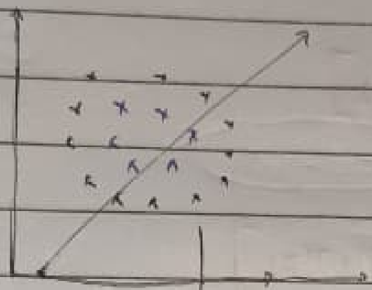
Hard Classifying SVM.

01/11/25

SVM Kernels Part - 3 !!

↓

Converts x dimention → x+n dimention
(lower) transform (higher)

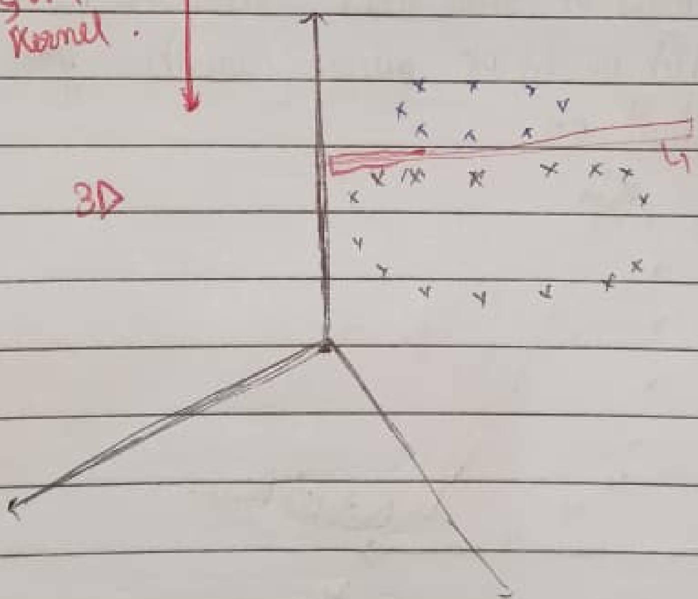


2D

↳ Not linearly separable. meaning, the features (x & x) are not in a form to be separated

Via SVM Kernel.

↓



3D

↳ separated

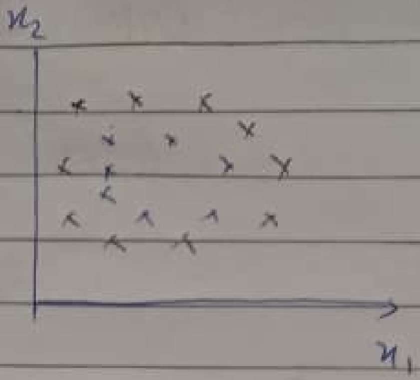
This dimensionality conversion occurs with help of mathematical formulas.

① Polynomial Kernels.

features to be used → $x_1, x_2, y_0, x_1^2, x_2^2, x_1 x_2$

y_0 → o/p feature

$$f(x_1, x_2) = (x_1^T \cdot x_2 + 1)^d$$



$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \cdot \begin{bmatrix} x_1 & x_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1^2 & x_1 x_2 \\ x_2 x_1 & x_2^2 \end{bmatrix}$$

$x_2 x_1$ a $x_1 x_2$.

though we initially used x_1 & x_2. we could've also used $x_1^2, x_2^2, x_1 x_2$ which would've given us the final result as